Life Checkpoint: Circuits + Python - Teacher Key

Mark each step independently. Continue after an incorrect earlier answer.

MARKING RULE: Award marks for correct evidence in each box even when an earlier answer is wrong. Accept equivalent current directions and signed equations when the student's work is internally consistent.

A. Ohm's law and power (8 marks)

```
2.0 kΩ = 2000 Ω

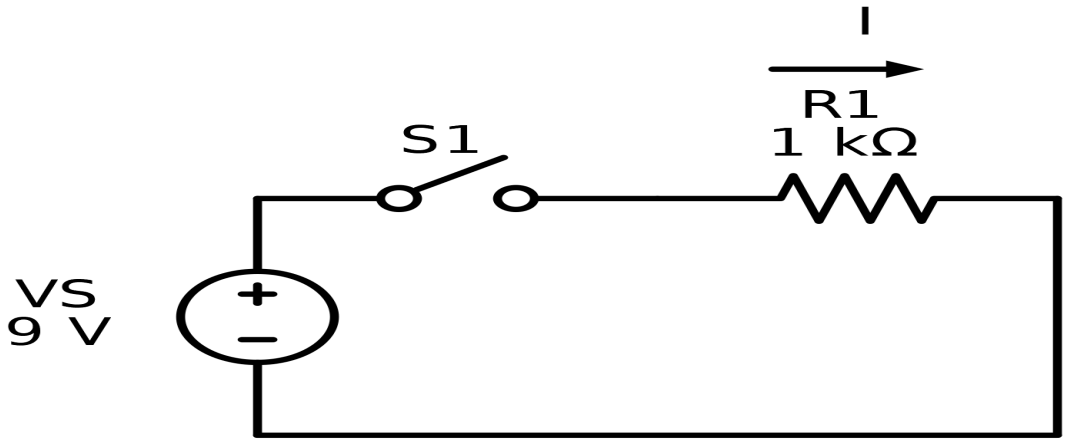
I = V / R = 12 / 2000 = 0.006 A = 6 mA

P = V x I = 12 x 0.006 = 0.072 W = 72 mW

Twice the power is 0.144 W, so choose 1/4 W.
```

Marking: 1 conversion, 1 formula, 2 current, 1 power formula, 2 power, 1 rating.

B. Draw a simple schematic (4 marks)



Marking: 1 complete loop, 1 correct symbols, 1 values/polarity, 1 current arrow. Equivalent clean layouts earn full marks.

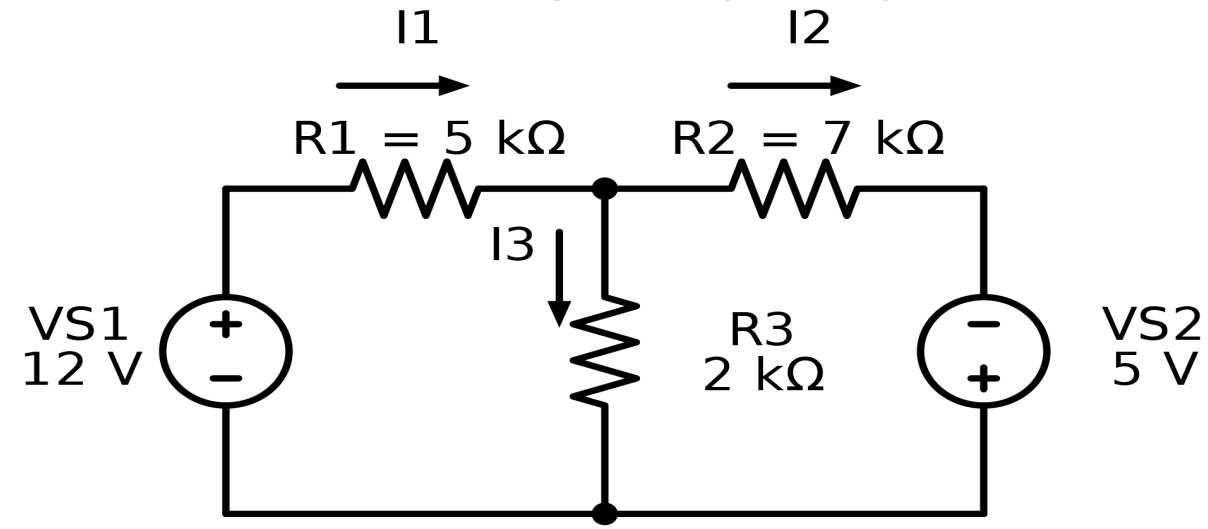
PAGE 1 TOTAL: 12 marks

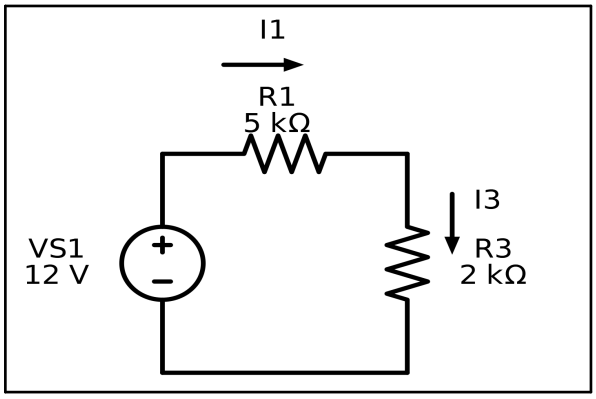
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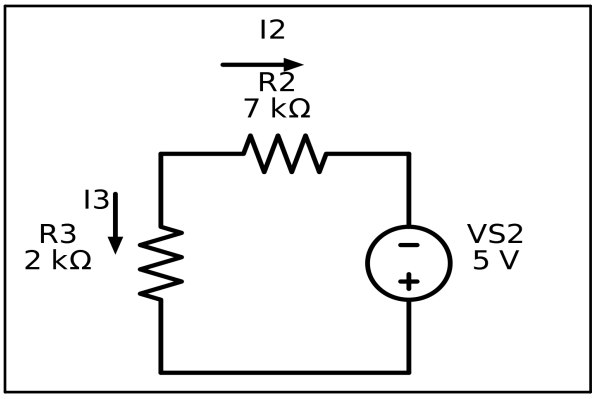
Mark each step independently. Continue after an incorrect earlier answer.

C. Read and redraw the two-loop circuit (8 marks)





Loop 1: VS1, R1, R3 (3)



Loop 2: R3, R2, VS2 (3)

KCL at the centre node (2 marks)

```
I1 = I2 + I3

I3 = I1 - I2
```

Marking: Continue marking the equations even if one isolated-loop drawing is incomplete.

PAGE 2 TOTAL: 8 marks

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D. KVL, standard form, and solution (12 marks)

1. Loop 1 (4 marks)

```
+12 - 5I1 - 2I3 = 0

+12 - 5I1 - 2(I1 - I2) = 0

+12 - 7I1 + 2I2 = 0

7I1 - 2I2 = 12
```

2. Loop 2 (4 marks)

```
+5 - 7I2 + 2I3 = 0

+5 - 7I2 + 2(I1 - I2) = 0

+5 + 2I1 - 9I2 = 0

-2I1 + 9I2 = 5
```

3. Solve and check (4 marks)

```
I1 = 2 mA

I2 = 1 mA

I3 = 1 mA

KCL: 2 mA = 1 mA + 1 mA
```

E. Matrix bridge (4 separate marks)

Equation	I1 coefficient	I2 coefficient	Right side
Loop 1	7	-2	12
Loop 2	-2	9	5

```
[  7  -2 ] [ I1 ]   [ 12 ]

[ -2   9 ] [ I2 ] = [  5 ]
```

Marking: Record matrix evidence separately if matrix form has not yet been independently assessed.

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Mark each step independently. Continue after an incorrect earlier answer.

F. Python and NumPy (12 marks)

1. Ohm's-law program (4 marks)

```
voltage_V = 12

resistance_ohm = 2000

current_A = voltage_V / resistance_ohm

power_W = voltage_V * current_A

print(current_A)     # 0.006

print(power_W)       # 0.072
```

Marking: 1 mark per correct variable in the two formulas; accept equivalent numeric expressions.

2. NumPy array (4 marks)

```
import numpy as np

voltages_V = np.array([2, 4, 6, 8])

resistance_ohm = 2000

currents_mA = voltages_V / resistance_ohm * 1000

print(currents_mA)   # [1.0 2.0 3.0 4.0]
```

Marking: 1 array constructor, 1 resistance variable, 1 calculation, 1 output.

3. Repair the type error (4 marks)

```
resistance_ohm = 2000

current_A = voltage_V / resistance_ohm

print(current_A)     # 0.006
```

Marking: 1 identifies quoted number, 2 corrected assignment syntax, 1 correct output.

MARK RECORD: Circuits ___/32 Python ___/12 Core ___/44 Matrix ___/4

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